

The Knowledge Lifecycle of Large Language Models

A model holds two memories. When they disagree, training usually wins.

Acquire

Store

Retrieve

Update

Forget



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Start with a question

You paste a document into a model's prompt.

The document contradicts what the model learned.

Which one wins?

Most of us assume the prompt wins. That assumption is what this talk tests.

The field is fragmented

- Five research communities each own one part of how a model handles a fact:
 - retrieval-augmented generation · knowledge editing
 - continual learning · context engineering · agent memory
- Each has its own benchmarks. Each passes them.
- A deployed system can pass all five and still fail, because nothing tests what happens **between** them.

Across 20 representative papers, median coverage is **two** of five stages.

The framework: five stages

Stage	What happens to the fact	Studied as
■ Acquire	Training compresses a corpus into weights	Pre-training, fine-tuning
■ Store	Weights, an external index, or both	Parametric memory, vector stores
■ Retrieve	Attention recalls, or a pipeline fetches	Retrieval-augmented generation
■ Update	The world changes; copies must follow	Knowledge editing, continual learning
■ Forget	Removed on purpose, or lost by accident	Unlearning, catastrophic forgetting

The failure in this talk sits between **Retrieve** and **Update**.

A documented case: Vioxx

- Rofecoxib, sold as Vioxx, was approved in 1999 as a COX-2 inhibitor.
- Withdrawn worldwide in **September 2004** after trials showed raised risk of heart attack and stroke.
- A model trained on pre-2004 text holds “approved”. The correction exists, and can be retrieved.



The measurement

	Without the notice	With the notice in the prompt
Answers “safe”	37.53%	42.58%
Answers “withdrawn”	0.0004%	0.0006%
Most likely word	“safe”	“safe”

- The withdrawal notice is **in the prompt**, and confidence in “safe” **rises**.
- The correct answer is left at roughly **six chances in a million**.
- Retrieval succeeded. Resolution failed.

This is not hallucination



No benchmark for retrieval, editing, or memory tests that second step,
because it belongs to none of them.

The metric: Lifecycle Desynchronization

$$\hat{\mathcal{D}}_{sync} = -\ln P(y^* \mid q, \theta_{stale}, \mathcal{M}_{now})$$

- The surprisal of the **correct** answer while the corrective document is present.
- Measured in nats: n nats means the right answer holds probability e^{-n} .
- Above 9.2 nats, no realistic decoding recovers it.



Two numbers, not one

	$\mathcal{I}_{ctx}$ low	$\mathcal{I}_{ctx}$ high
$\mathcal{D}_{sync}$ low	fact never conflicted	context resolved the conflict
$\mathcal{D}_{sync}$ high	resolution failure	drift: influential but losing

- $\mathcal{I}_{ctx}$ measures whether the document moved the model **at all**.
- On Vioxx it is **0.033** nats. The document is **inert**.
- That pairing is what separates a broken retriever from a model that ignores what it retrieved.

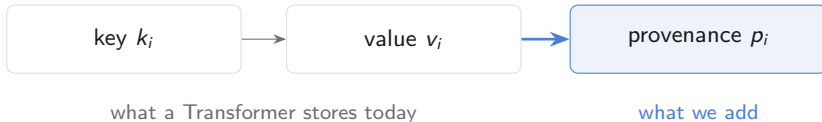
Three cases, three regimes

Fact change	What the model does	Regime
Vioxx withdrawn, 2004	ignores the notice entirely	resolution failure
Elizabeth II died, 2022	the notice boosts “Queen”	the exposure trap
Twitter renamed X, 2023	moves hard, still answers “Twitter”	drift

The middle row is the uncomfortable one: **a correct document made the wrong answer stronger.**

The cause: storage discards provenance

- Training teaches a model **what** is true.
- It never records **when** the fact was learned, or **where** it came from.
- So at inference there is no principled basis for preferring fresh evidence over a confident old memory.



The proposal: gate stale facts at inference

$$\text{FFN}_{\text{steered}}(x) = \sum_{i=1}^{d_m} S(\Delta\tau_i) f(x \cdot k_i) v_i, \quad S(\Delta\tau_i) = \sigma(\gamma - \beta \cdot \Delta\tau_i)$$

- Fresh facts pass with $S \approx 1$; stale facts are attenuated toward 0.
- **The weights are never modified**, so the specificity bottleneck that limits static editing does not apply.
- Cost: about 1M parameters on a 7B model, under **0.02%**.

Think of it as a volume control on every stored fact, turned down by age.

What is measured, and what is proposed

Measured

The metric, on real models,
deterministically,
reproducible on a laptop

Proposed

The architecture,
with an idealized proof
under stated assumptions

The Provenance Vector has **not** been trained. It needs a pretraining run, and the paper says so in its limitations rather than leaving you to find out.

Check it yourself

- Every number in this talk comes from **one deterministic script**. No sampling, identical digits on any machine.
- The model is GPT-2 base, 124M parameters: small, fully open, and free of instruction tuning that would confound the result.
- The demonstration runs GPT-2 **inside your browser**. Nothing you type leaves your laptop.

Paper and code github.com/Amey-Thakur/LLM-KNOWLEDGE-LIFECYCLE

Live demonstration huggingface.co/spaces/ameythakur/llm-knowledge-lifecycle

What comes next

- **Does it survive scale? Yes.** Across six open models from 124M to 1.5B, not one answers the Vioxx probe correctly, and inside the GPT-2 family the measurement is not even monotone in size.
- **Does instruction tuning fix it? Not on this probe.** The tuned models resolve the Twitter and Monarch conflicts and still lose Vioxx; the largest answers “unsafe”, which is the right direction and the wrong token.
- **Does reasoning-mode inference help?** Extended test-time compute should, in principle, resolve conflicts the forward pass cannot.
- **Multimodal.** A text prior overriding a contradicting image is the same failure in another modality.

The first two are answered, and the answer to both is largely negative. Reporting that is the point.

In three sentences

- Five separate research fields are five stages of **one lifecycle**, and systems fail at the boundaries nobody benchmarks.
- Those failures are **measurable**: a model with the correction in its prompt still answers wrong, at 12.05 nats.
- The cause is that storage **discards provenance**, and an inference-time gate can address it without touching the weights.



Thank you

Questions



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github.com/Amey-Thakur/LLM-KNOWLEDGE-LIFECYCLE

Backup: anticipated questions

- **Why GPT-2 and not a frontier model?** Small, fully open, no instruction-tuning confound, and it makes the result checkable on a laptop. The claim is that the failure exists and is measurable, not that it is universal.
- **Is this just hallucination?** No. The retriever succeeded, and $\mathcal{I}_{ctx} = 0.033$ proves the context was present and inert.
- **Does it disappear at scale?** Unknown, and that is the honest answer. The sweep is running; we will report what it shows.
- **Why not just edit the weights?** Editing damages neighboring facts, and degrades as edits accumulate. Gating leaves the weights untouched.
- **Is the proof meaningful if untrained?** It shows the mechanism is structurally sound given accurate provenance. The assumptions are stated, not hidden.